

Analogue Variable Speed Drive [VSD] Motor MAPS Template (Electrical)

The Analogue Variable Speed Drive (VSD) Motor MAPS template is designed to control a motor that is connected to a VSD with Analogue Set point and feedback.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This document describes the Analogue VSD MAPS Template, which provides control over the VSD motor with Analogue Speed set point and feedback. For details of its required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

Analogue VSD Motor Functional Philosophy

The VSD start output is enabled once all interlocks are ready and a start request is received.

Field Interlocks

- MCC Healthy Feedback
- Safety Interlock Feedback

Interlock fault recovery process – once a field interlock is activated the faceplate will display a red indication, if the interlock is recovered and still latched in the PLC function block and the faceplate will show a green indication and the reset button on the faceplate will flash, pressing the Reset button will unlatch the interlock and display a grey indication and allow the DOL to become ready for operation.

Start and stop requests are initiated by selecting a mode on the VSD faceplate:

- **Auto mode** - the auto control program initiates a start and stop, normal PLC control.
- **Manual mode** - the operator can start and stop from the faceplate.

Typical Faceplate Graphics for the Analogue VSD Motor MAPS Template

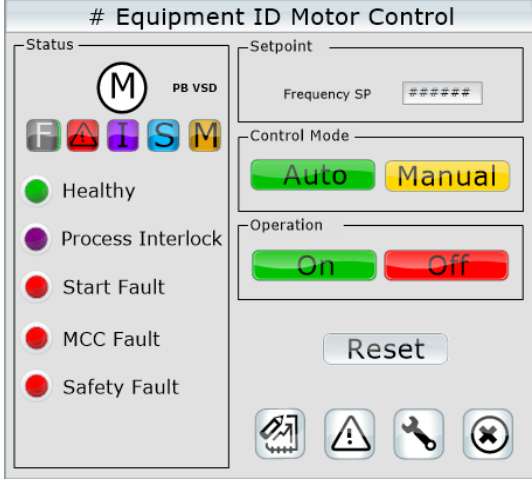
VSD ISO	VSD Aerator	VSD Standard Pump Side View Left	VSD PUMP LEFT	VSD PUMP RIGHT
0000_ABCDE_123 I F S M M AI VSD	0000_ABCDE_123 I F S M  AI VSD	0000_ABCDE_123 I F S M  AI VSD	0000_ABCDE_123 I F S M  AI VSD	0000_ABCDE_123 I F S M  000.00 Hz
I	Interlock			
H	Manual Mode			
S	Simulation Mode			
M	Maintenance Mode			
F	Local Field Control			
0000_ABCDE_123	Motor Control Name			
Graphics legend:				
Line colour	Fill colour	DOL status		
Black	Grey	Stopped		
Grey	White	Stopping		
Green	White	Starting		
Black	Green	Running		
Black	Yellow/Red	Fault Active		

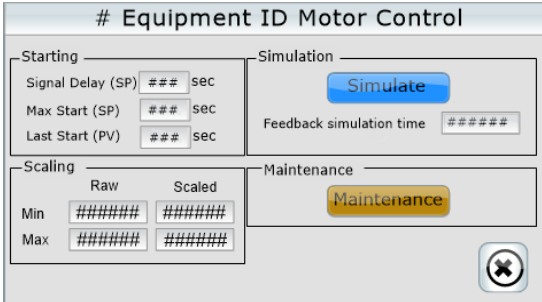
Signal Description		Agent Type	Basic
SCADA Control Words:			
SSCL 0	Control Word	Marshal	x
SSCL 1	Fail To Start Set point	Analog	x
SSCL 2	Running Set Point	Analog	x
SSCL 3	Simulation Set Point	Analog	x
SSCL 4	Frequency Set Point	Analog	x
SSCL 5	Minimum Raw Analog Input	Analog	x
SSCL 6	Maximum Raw Analog Input	Analog	x
SSCL 7	Minimum Scaled Analog Output	Analog	x
SSCL 8	Maximum Scaled Analog Output	Analog	x

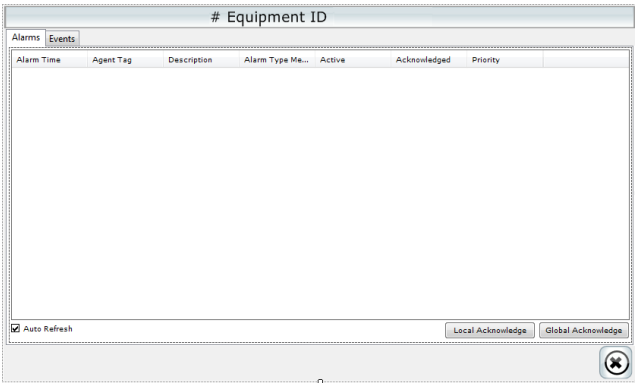
Signal Description		Agent Type	Basic
SCADA Status Words:			
SSSL 0	Status Word	Marshal	x
SSSL 1	Starting Time Feedback	Analog	x
SSSL 2	Frequency Feedback	Analog	X

Digital Inputs:		Basic
DI 0	Run Feedback (On = Running)	x
DI 1	MCC Healthy (On = Healthy)	x
DI 2	Safety Interlock (On = Healthy)	x
DI 3	Local(Field) / Remote(Desk) Selection (On = Local)	x
Digital Outputs:		
DO 0	Device Start	x
Analogue Inputs:		
AI 1	Frequency Feedback	x
Analogue Outputs		
AO 1	Frequency Set Point	x

SCADA Control Word (Adroit Marshal Agent)		Event Logged	Basic
Bit 0	Auto/Manual Select		x
Bit 1	Spare		x
Bit 2	Maintenance Select		x
Bit 3	Simulation Select		x
Bit 4	Alarms All Reset		x
Bit 5	Spare		
Bit 6	Manual Start Request		x
Bit 7	Spare		
Bit 8	Manual Stop Request		x
Bit 9	Spare		
Bit 10	Spare		
Bit 11	Spare		
Bit 12	Spare		
Bit 13	Spare		
Bit 14	Spare		
Bit 15	Spare		

Template Graphic: Home Screen	Control	Descriptions
	On	Manually start the VSD
	Off	Manually stop the VSD
	Reset	Reset the VSD after a fault has been repaired
	Healthy	Healthy to start the VSD
	Auto	VSD is started from automatically through the
	Manual	VSD can be started manually from the buttons on
	I	Interlock
	H	Manual Mode
	S	Simulation Mode
	M	Maintenance Mode
	F	Local Field Control
	A	Latched Fault
	Trend	Auto Mode
	Alarm	Open the Trend Window
	Setup	Opens the Alarm and Events window
	Help	Opens the Setup Screen

Template Graphic: Setup Screen	Control	Descriptions
	Simulate	Enable simulation of VSD, disables the physical
	Maintenance	Enable maintenance mode (Disables all
	Start Signal	Delay the PLC to accept the VSD is running
	Max Start (SP)	Time for running signal is on after start signal is
	Last Start (PV)	Last time for running signal was on after start
	Min, Max Raw	Raw speed input values for scaling
	Min, Max Scaled	Engineering speed values for scaling
	Feedback Time	Time for simulation to simulate run feedback

Template Graphic: Alarms & Events Screen	Control	Descriptions
	Alarms	Display all the alarms filtered for the current Digital Output
	Events	Display all the events filtered for the current Digital Output.
	Local Acknowledge	Acknowledge all the alarms on locally on the current machine
	Global Acknowledge	Acknowledge all the alarms on globally on all the machines.